THE NATIONAL INSTITUTE OF ENGINEERING, MYSURU

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

MACHINE LEARNING MINI-PROJECT

BANKNOTE AUTHENTICATION USING LOGISTIC REGRESSION

Submitted by

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1. INTRODUCTION

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions.

This project focuses on Banknote Authentication using Logistic Regression, a supervised machine learning algorithm used for binary classification.

The objective is to classify banknote records into two categories using four numerical features extracted from scanned banknote images. The model is trained using the training dataset and evaluated using unseen testing records.

2. PROBLEM STATEMENT

The objective of this project is to develop and evaluate a Logistic Regression model that predicts the class of banknotes based on four numerical measurements.

The model is trained using labelled data and tested using unseen records. Its performance is evaluated using different classification metrics to understand its strengths and limitations.

3. DATASET DESCRIPTION

The Banknote Authentication dataset was obtained from the UCI Machine Learning Repository.

Dataset details:

Description	Value
Total records	1,372
Total columns	5
Input features	4
Target variable	Class
Missing values	0
Duplicate records	24

Input features:

1. Variance – Variance of the wavelet-transformed image.
2. Skewness – Skewness of the transformed image.
3. Curtosis – Kurtosis-related measurement.
4. Entropy – Entropy of the image.

The target variable is Class, which contains two numerical labels, 0 and 1.

The official UCI dataset page describes the genuine and forged specimens but does not explicitly document the numerical mapping of the two class labels. Therefore, the original labels have been retained without making an unsupported assumption about their meanings.

4. EXPLORATORY DATA ANALYSIS

The dataset was explored using Pandas and NumPy.

The analysis included:

- Displaying the first ten records.
- Checking the number of rows and columns.
- Checking the column names and data types.
- Checking missing values.
- Checking duplicate records.
- Calculating descriptive statistics.
- Analysing class distribution.

The dataset contains 1,372 records and five columns. All four input features have the float64 data type, while the target Class has the integer data type.

The missing-value check showed zero missing values in all columns.

The duplicate-record check identified 24 duplicate records. These records were retained without removal.

Class distribution:

Class	**Number of records**	**Percentage**
Class 0	762	55.54%
Class 1	610	44.46%
Total	1,372	100%

The dataset is relatively balanced because the difference between the two class percentages is not very large.

Descriptive statistics were generated to examine the count, mean, standard deviation, minimum, maximum and quartiles of the features. The four input features have different numerical ranges, indicating that feature scaling is useful.

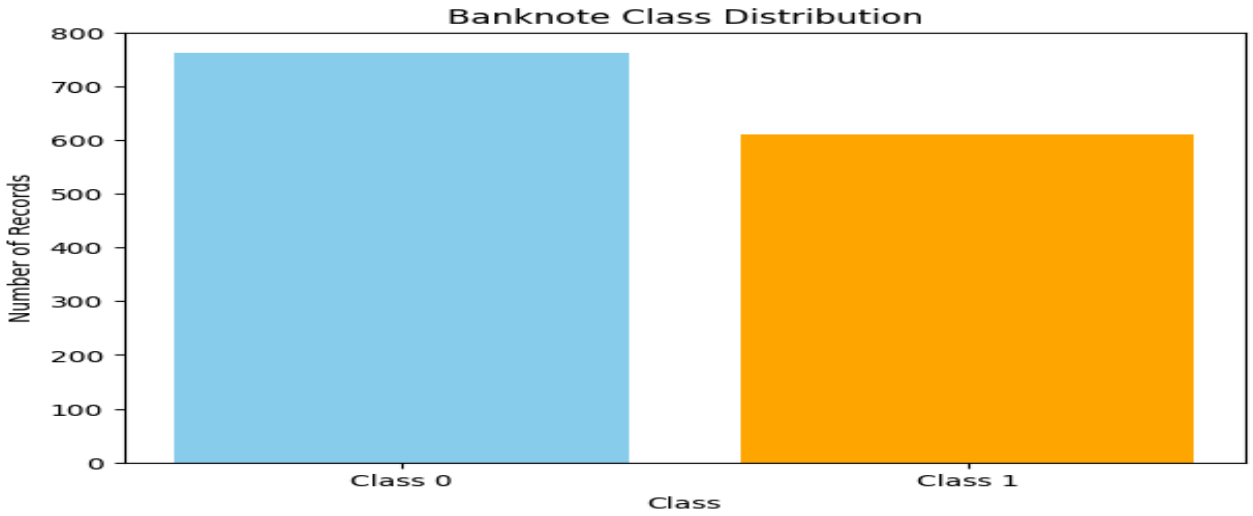
5. DATA VISUALISATION

The following visualisations were created using Matplotlib:

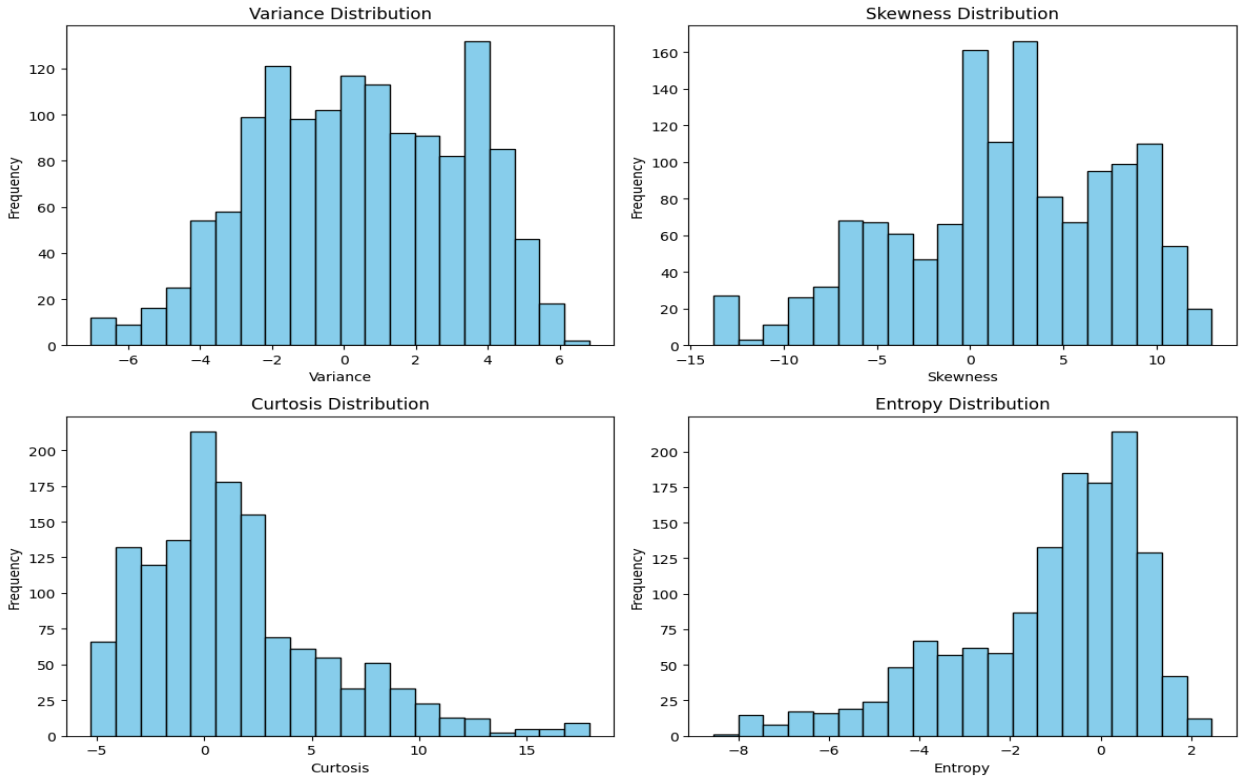
1. Bar chart showing the distribution of Class 0 and Class 1.
2. Histogram of Variance.
3. Histogram of Skewness.
4. Histogram of Curtosis.
5. Histogram of Entropy.
6. Scatter plot of Variance versus Skewness.
7. Scatter plot of Curtosis versus Entropy.

Observations:

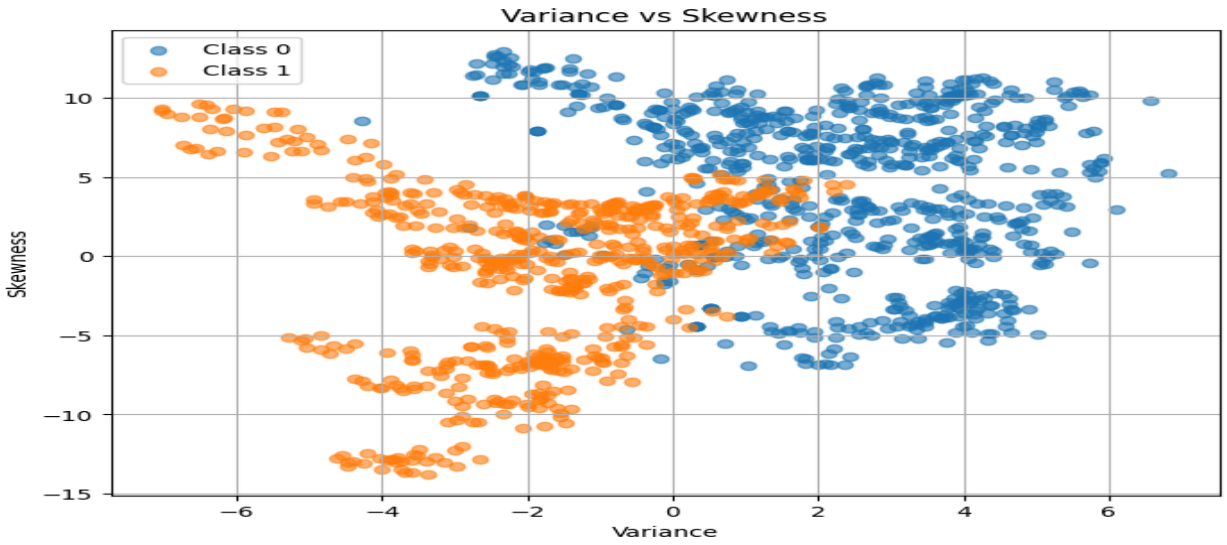
- The bar chart shows that Class 0 has 762 records, while Class 1 has 610 records.



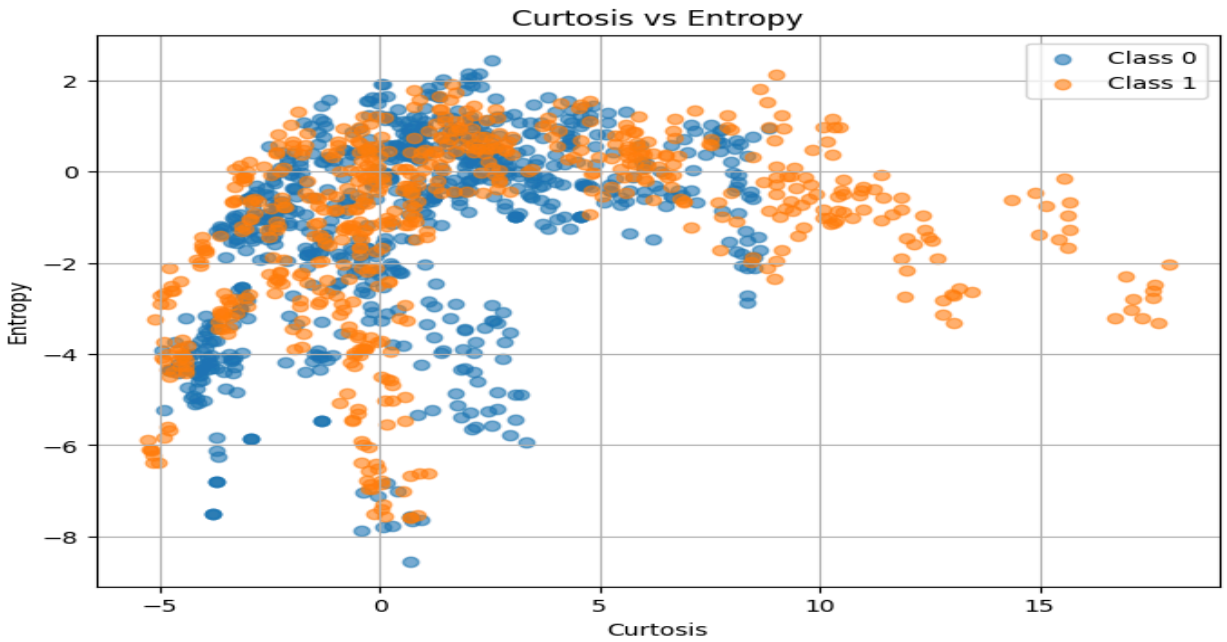
- The histograms show that the four features have different numerical distributions and ranges.



- The Variance versus Skewness scatter plot shows differences in the distribution of the two classes, although some overlapping regions are present.



- The Curtosis versus Entropy scatter plot also shows differences between the two classes.



- Combining all four features may help the model distinguish between the two classes more effectively.

6. METHODOLOGY

6.1 Feature and target separation

The four input features were separated from the target variable.

Input features (X):

- Variance
- Skewness
- Curtosis
- Entropy

Target variable (y):

- Class

The input matrix contains four features, and the target contains one class label for each record.

6.2 Train-test splitting

The dataset was divided into 80% training data and 20% testing data.

The splitting process used random_state=42 and stratify=y to preserve approximately the same class proportions in both subsets.

Dataset	Record
Training data	1,097
Testing data	275
Total	1,372

Training data was used to train the model, while testing data was reserved for evaluating its performance.

Stratified splitting helps maintain the class distribution in both subsets.

6.3 Feature scaling

StandardScaler was used for feature scaling.

The scaler was fitted only on the training data and then used to transform both training and testing data.

Feature scaling places the numerical features on comparable scales, which is useful for Logistic Regression.

Data leakage occurs when information from the testing dataset is used during model training or preprocessing. Fitting the scaler before splitting the data could cause data leakage and make the evaluation unreliable.

6.4 Logistic Regression model

Logistic Regression from Scikit-learn was used for binary classification.

The model was created using the following settings:

Parameter	Value
Algorithm	Logistic Regression
Maximum iterations	1000
Regularisation	L2
Solver	lbfgs
C	1.0

The model was trained successfully using the scaled training features and their corresponding class labels.

Logistic Regression is suitable for binary classification because it estimates the probability of a record belonging to a particular class.

The sigmoid function converts the model score into a probability between 0 and 1.

7. MODEL PREDICTION AND EVALUATION

The trained Logistic Regression model was used to predict the class labels of the 275 testing records.

Predicted probabilities for both Class 0 and Class 1 were also generated.

A comparison of actual and predicted class labels was displayed for at least 15 test records. Probability results were also displayed for at least ten records.

A predicted probability of 0.80 for Class 1 means that the model estimates an 80% probability of that record belonging to Class 1. It does not guarantee that the prediction is correct.

7.1 Evaluation metrics

The following evaluation results were obtained:

Metric	Result
Accuracy	0.9709
Accuracy percentage	97.09%
Precision for Class 1	0.9385
Recall for Class 1	1.0000
F1-score for Class 1	0.9683

Class 1 was treated as the positive class.

Classification report:

Class	Precision	Recall	F1-score	Support
0	1.00	0.95	0.97	153
1	0.94	1.00	0.97	122
Accuracy			0.97	275
Macro average	0.97	0.97	0.97	275
Weighted average	0.97	0.97	0.97	275

Interpretation:

Accuracy measures the proportion of correct predictions among all testing records.

Precision measures how many predicted Class 1 records actually belong to Class 1.

Recall measures how many actual Class 1 records were correctly identified.

The F1-score combines precision and recall into a single measure.

The model achieved 97.09% accuracy, indicating strong classification performance on the test set.

Accuracy alone does not provide a complete picture because it does not identify the types of errors made by the classifier. Therefore, precision, recall, F1-score and the confusion matrix are also important.

8. CONFUSION MATRIX

The confusion matrix was generated using the test predictions, with labels ordered as [0, 1].

Numerical confusion matrix:

Actual / Predicted	**Class 0**	**Class 1**
Class 0	145	8
Class 1	0	122

The confusion matrix values are:

Term	**Value**
True Negative (TN)	145
False Positive (FP)	8
False Negative (FN)	0
True Positive (TP)	122

Interpretation:

- True Negative: 145 actual Class 0 records were correctly predicted as Class 0.
- False Positive: 8 actual Class 0 records were incorrectly predicted as Class 1.
- False Negative: No actual Class 1 records were incorrectly predicted as Class 0.
- True Positive: 122 actual Class 1 records were correctly predicted as Class 1.

The model correctly classified 267 out of 275 testing records.

If Class 1 is confirmed to represent forged notes, recall would be especially important because it measures the ability to identify forged notes. The class mapping must be verified before assigning genuine or forged meanings to these errors.

9.CLASSIFICATION THRESHOLD COMPARISON

Three classification thresholds were investigated:

- 0.30
- 0.50
- 0.70

The threshold determines the minimum predicted probability required for a record to be classified as Class 1.

At a threshold of 0.30, the model uses a lower probability requirement for predicting Class 1.

At a threshold of 0.50, the standard classification threshold is used.

At a threshold of 0.70, the model requires a higher probability before predicting Class 1.

Threshold comparison results:

The accuracy, precision, recall and F1-score were calculated for each threshold using the same test probabilities.

Insert the actual threshold comparison table and the three confusion matrix visualisations generated by Google Colab in this section.

Changing the threshold affects the number of positive predictions and the balance between precision and recall. A lower threshold generally increases positive predictions, while a higher threshold generally reduces them.

The most suitable threshold depends on the cost of false positive and false negative errors rather than accuracy alone.

11. LIMITATIONS

1. The dataset contains measurements from a particular collection of banknote specimens and may not represent all currencies or scanning conditions.
2. Logistic Regression may not capture every complex nonlinear relationship between the features.
3. The dataset contains 24 duplicate records, which were retained. Some duplicates could occur in both training and testing subsets, potentially making the evaluation optimistic.
4. Good performance on this dataset does not guarantee reliable performance on new banknotes or real banking systems.

Therefore, the model is suitable for educational experimentation but is not certified for real banking, legal or security decisions.

12. CONCLUSION

This mini-project focused on Banknote Authentication using Logistic Regression, a supervised Machine Learning algorithm for binary classification. The UCI Banknote Authentication dataset containing 1,372 records and four numerical input features, namely Variance, Skewness, Curtosis and Entropy, was used to develop the classification model.

The dataset was explored to understand its structure, class distribution, missing values and duplicate records. Data visualisations were created using bar charts, histograms and scatter plots. The dataset was divided into 80% training data and 20% testing data using stratified splitting. StandardScaler was applied to standardise the features without causing data leakage.

The Logistic Regression model achieved an accuracy of 97.09%, with a precision of 93.85%, recall of 100% and F1-score of 96.83% for Class 1. Its performance was also evaluated using a confusion matrix. Different classification thresholds of 0.30, 0.50 and 0.70 were investigated to understand their effects on predictions.

Recall is particularly important when failing to identify forged banknotes is costly, provided the forged-note class is correctly identified. However, duplicate records and the dataset's limited scope are important limitations.

Overall, this project provided practical knowledge of data preprocessing, classification, model evaluation and threshold selection.

13. REFERENCES

1. Lohweg, V. (2012). *Banknote Authentication Dataset*. UCI Machine Learning Repository. https://doi.org/10.24432/C55P57
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